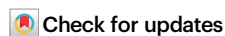


Structures of partially occupied hetero-tetramers provide insight into kainate receptor activation and desensitization

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Kainate receptors (KARs) are critical mediators and modulators of synaptic transmission which undergo rapid activation and desensitization upon binding of the neurotransmitter glutamate. Under various physiological and pharmacological conditions agonist binding likely occurs to only a subset of subunits within these tetrameric receptors, motivating an analysis of the functional and conformational effects of partial versus complete ligand occupancy. Here we report cryo-EM structures of the GluK2/GluK5 hetero-tetramer under partially-occupied conditions using 5-iodowillardiine and AMPA as GluK5-selective agonists. High-resolution pre-active state structures containing closed/open ligand binding domain (LBD) dimers with intact interfaces reveal gating-associated interface reshaping, inter-dimer motions, and pore-linker repositioning in response to asymmetric agonist binding. Interfacial LBD mutations to a central cluster formed by the GluK5 subunits and to an inter-dimer interface between GluK2 and GluK5 subunits, highlight the roles of interactions between LBD dimers in controlling receptor function, including the distinct slow deactivation of GluK5-containing receptors. Finally, the absence or presence of intact, partially, and fully ruptured LBD interfaces under different ligand conditions allows us to propose a revised model of stepwise ionotropic glutamate receptor activation and desensitization.

Throughout the nervous system, a diverse ensemble of metabotropic (mGluRs) and ionotropic (iGluRs) glutamate receptors respond to the neurotransmitter glutamate to sculpt synaptic transmission and plasticity¹. The three major iGluR subfamilies, namely AMPA (AMPA; GluA1-4), kainate (KARs; GluK1-5), and NMDA (NMDARs; GluN1-3) receptors, show substantial functional variety². In all three subfamilies receptor diversity is increased by formation of either homo- or hetero-tetrameric complexes, in combination with other mechanisms, such as alternative splicing, posttranslational modifications, and association with auxiliary subunits^{2,3}. Unlike AMPARs

and NMDARs, which drive the bulk of synaptic transmission, KARs are thought to play more specialized, modulatory functions in both the pre- and post-synaptic compartments where they control different forms of plasticity^{4,5}. In addition, KARs play roles in the peripheral nervous system, including in pain, cold sensing, and inflammation⁶⁻⁸. Based on these physiological roles, as well as their restricted expression profiles^{9,10} and unique binding pockets¹¹⁻¹³, KAR subtypes are increasingly recognized as attractive drug targets for a range of disorders including epilepsy, pain, ischemic conditions, and psychiatric disorders^{4,14}.

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Following exposure to glutamate or synthetic agonists, both AMPARs and KARs rapidly activate and desensitize. The molecular diversity of AMPARs and KARs leads to a range of time scales for glutamate-driven activation (<1 ms), desensitization (<10 ms), deactivation (1–100 ms), and recovery from desensitization (50 ms to 10 s). For example, KARs typically show ~50-to-1000-fold slower recovery from desensitization than AMPARs^{15–22} and GluK5-containing KARs show an unusually slow deactivation^{18,23,24}.

Critically, not all iGluRs may be fully occupied by glutamate during brief synaptic release events^{25–27}. This may result in partial activation, as single channel studies have shown that AMPAR conductance scales with glutamate occupancy with binding to one or two subunits being sufficient to produce only partial activation^{28–30}. Less is known about the occupancy-dependence of iGluR desensitization, but various models have been proposed including those where a single agonist is sufficient to drive desensitization^{18,31,32}. Incomplete ligand occupancies might be particularly relevant for heteromeric receptors incorporating subunits with different glutamate affinities. A notable example are heteromeric KARs composed of ‘low-affinity’ subunits (GluK1–3) and ‘high-affinity’ subunits (GluK4–5). Indeed, GluK2/GluK5 heteromers are thought to be one of the most prevalent KARs in the central nervous system^{33,34}, and recent data show a wide potential for other di-, tri-, and tetra-heteromeric KAR combinations¹⁰. Subtype-specific synthetic ligands or agonists with different dissociation kinetics can also result in partially occupied iGluRs. For example, specific liganding of GluK2 or GluK5 subunits within a GluK2/K5 heteromer produces non-desensitizing currents^{35–38} and subunit-specific blockade can potentiate KAR and AMPAR currents via impairment of desensitization²⁴. These findings reveal that iGluR agonist partial occupancy states exist, which can trigger activation but not desensitization. However, despite the physiological and pharmacological significance of partially occupied states, the molecular basis for the relationship between ligand occupancy and activation and desensitization of iGluRs is not fully deciphered.

Structurally, iGluRs form a layered domain architecture with extracellular pairs of dimeric amino-terminal domains (ATDs) followed by dimeric but domain-swapped ligand-binding domains (LBDs), a common transmembrane domain (TMD) pore region, and intracellular carboxy-terminal regions^{2,39}. Great progress has been made in understanding the structural basis of iGluR activation and desensitization, with the majority of work on AMPARs and NMDARs. A consensus understanding is that orthosteric agonist binding drives both LBD clamshell closure and LBD dimer reorganization to reposition linkers, which pull on the TMD to enable pore opening. On overlapping time scales, LBD dimers loosen or rupture, relieving TMD linker tension, resulting in pore closure and functional desensitization. However, it is unknown how general this mechanism is and how it may vary between iGluR subclasses and subtypes. For example, in KARs, fully ruptured and partially ruptured states have been observed, but their relationship to desensitization and ligand occupancy remains unclear^{10,23,40–47}. Moreover, the rapid onset of desensitization following agonist binding has made it difficult to capture active states with fully liganded receptors typically being found in inactive states. Structural studies aimed at non-desensitized agonist-bound states have thus used positive allosteric modulators (PAMs) or desensitization blockers^{30,43,48–52}. Notably, a very recent study used physiological temperature treatment prior to sample freezing to capture active state structures of fully liganded homotetrameric GluA2 AMPARs in the absence of PAMs or desensitization blockers, demonstrating distinct conformational effects of orthosteric versus allosteric ligands⁵³. In addition, the asymmetric arrangement of the LBD dimers raises the question of how distinct subunit positions (i.e., A/C pore-proximal versus B/D pore-distal subunits) contribute to channel gating. Based on their tighter pore-linker configurations, a gating dominance of B/D subunits has been suggested^{39,54}, but there is a paucity of structural information on

receptors partially occupied in specific positions. Notably, our prior structural studies showed that GluK5 takes the A/C positions in full-length GluK2/GluK5 heterotetramers²³. Owing to the higher glutamate affinity of GluK5 subunits compared to GluK2 subunits, positions A and C are more likely to be occupied under physiological conditions than positions B and D, yet it has been shown that GluK5 subunits are sufficient to produce activation when coexpressed with GluK2 variants with strongly lowered glutamate affinity³⁷. Indeed, even GluK5-selective agonists can activate these heteromers^{35,36}, which raises the question of how A/C agonist occupancy leads to gating-associated structural changes.

Here, we first determined cryo-EM structures of GluK2/K5 heterotetramers in the presence of an orthosteric ligand combination that fails to elicit desensitization. A high-resolution structure containing closed/open LBD dimers with intact interfaces reveals gating-associated interface reshaping, inter-dimer motions and pore-linker repositioning that likely reflect a pre-active state. Interfacial LBD mutations reveal major contributions of inter-dimer LBD interactions in controlling KAR gating. We also determined GluK2/K5 structures in the presence of AMPA, a ligand which we find produces partial desensitization and a range of intact, partially-ruptured, and fully-ruptured LBD layers, allowing us to propose a working model of agonist occupancy-dependent iGluR activation and desensitization.

Results

Cryo-EM structures of GluK2/GluK5 heterotetramers under non-desensitizing ligand conditions

We first performed patch-clamp recordings of agonist-evoked currents in HEK293 cells expressing either GluK2 homotetramers or GluK2/GluK5 heterotetramers. In both cases, rapid application of glutamate produced inward currents with rapid onset (<1 ms) followed by complete and fast (3–5 ms) desensitization (Figs. 1A; S1). Application of the GluK5-specific agonist 5-iodowillardiine (5-IW)^{36,55} to the same cells, including for extended time periods of up to 10 s, did not elicit any measurable currents from GluK2 homotetramers but produced substantial, non-desensitizing currents at GluK2/GluK5 heterotetramers (Figs. 1A; S1). This result is consistent with previous work suggesting that agonist occupancy of just two subunits of the same subtype within a heterotetramer can drive partial activation without triggering desensitization^{24,35}.

To probe the structural basis of these non-desensitizing currents, and as a potential means to visualize activated and intermediate states, we purified GluK2/GluK5 heterotetramers using our previously established protocol²³ and incubated them with 5-IW to mimic the non-desensitizing state observed functionally. In addition, samples were pre-incubated with the GluK2-specific antagonist kynurenic acid to avoid binding of any ambient glutamate from purification. We froze grids within 1–2 min of 5-IW application and determined cryo-EM structures (Fig. 1B). Two distinct states were observed with similar amounts of particles populating each class (class 1 = 208,265 particles; class 2 = 206,013 particles) (Fig. S2). For each state, particle subtraction and local refinement were performed on isolated ATDs and LBD-TMD modules. In both classes, higher resolution was observed at the ATDs (class 1 = 3.0 Å; class 2 = 2.9 Å) compared to the LBD-TMDs (class 1 = 3.5 Å; class 2 = 3.8 Å) (Fig. S2B–I). As previously, we identified GluK2 versus GluK5 subunits based on distinct glycosylation sites, revealing a 2:2 stoichiometry with GluK5 subunits in A and C positions, and a domain-swapped architecture with different subunit pairing in the ATD (A/B and C/D heterodimers) and LBD (A/D and B/C heterodimers) layers (Fig. 1C). Clear ligand densities were seen in both GluK2 and GluK5 LBDs, but the resolution was not sufficient to confidently model their binding pose (Fig. S2L–M).

We built models for class 1 and class 2 states (Table S1) and performed global comparisons with prior GluK2/GluK5 structures in antagonist-bound inactive or agonist-bound desensitized states

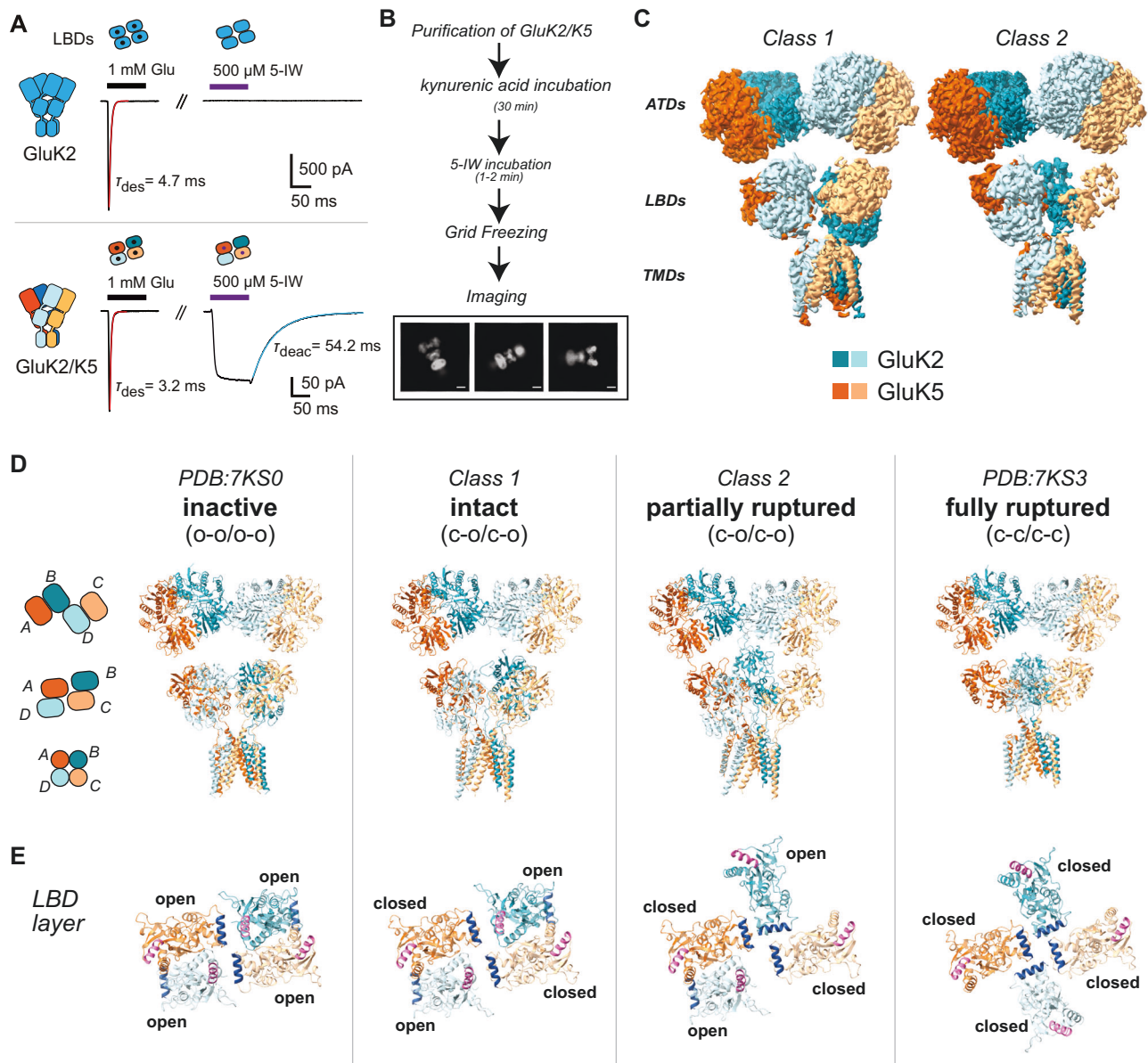


Fig. 1 | Cryo-EM structures of partially occupied GluK2/GluK5 heterotetramers.

A Patch-clamp recording of GluK2 homomers and GluK2/GluK5 heteromers expressed in HEK293 cells. Glutamate induces fully desensitizing currents in both cases (left). 5-IW, a GluK5-selective agonist (right), only activates GluK2/GluK5 heteromers, where it results in non-desensitizing currents. For details see Fig. S1. **B** Cryo-EM workflow and representative 2-dimensional class averages (bottom) for

partially occupied GluK2/GluK5 heteromers. **C** Full-length cryo-EM maps for the GluK2/GluK5 heteromer classes identified. **D** Full-length GluK2/GluK5 heteromer structural models for inactive (PDB 7KS0), intact (class 1, PDB 9OSF), partially ruptured (class 2, PDB 9OSG), and fully ruptured (PDB 7KS3) states. c=closed, o=open. **E** Top views of the LBD layer for the same structures as in D. Helix G is highlighted in blue, and helix B is highlighted in magenta.

(Fig. 1D). All four structures showed highly similar ATD conformations (Figs. 1D; S3) but distinct LBD conformations (Fig. 1E). We previously established that antagonist-bound GluK2/GluK5 represents an inactive state with all four LBDs containing open clamshells and intact dimer interfaces, whereas glutamate-bound GluK2/GluK5 contains all four LBDs in closed states with fully ruptured dimer interfaces enabling the formation of a desensitization ring, consistent with a fully desensitized state²³. In line with our ligand conditions, both of our additional structures contain a mix of closed GluK5 LBDs and open GluK2 LBDs (Fig. S4A–C). However, our class 1 state shows intact LBD dimers and thus represents an agonist-bound intact LBD state, which has not previously been observed across the KAR family, to the best of our knowledge, except in the presence of desensitization blockers⁴³. In our class 2 state, only one of the LBD dimers remains intact while the other has ruptured (Fig. 1E; S4D). This partially ruptured conformation is

reminiscent of what has been seen in some prior GluK structures^{41–43,45} and may represent an intermediate between active and fully desensitized states. Notably, we do not observe a fully ruptured LBD assembly under these ligand conditions, consistent with the lack of desensitization observed in response to 5-IW (Fig. 1A).

Analysis of LBD dimer conformational changes

We first focused our analysis on the LBDs, as previous structural and functional work has indicated that structural rearrangements of the LBD dimer interfaces are related to both activation and desensitization². We first focused on structural changes observed at the level of GluK2/GluK5 LBD dimers in the intact class 1 state. Compared to the antagonist-bound structures (Fig. 2A), the intact class 1 structure shows rearrangement of the upper D1 and lower D2 subdomain interfaces with an overall loosening of both interfaces

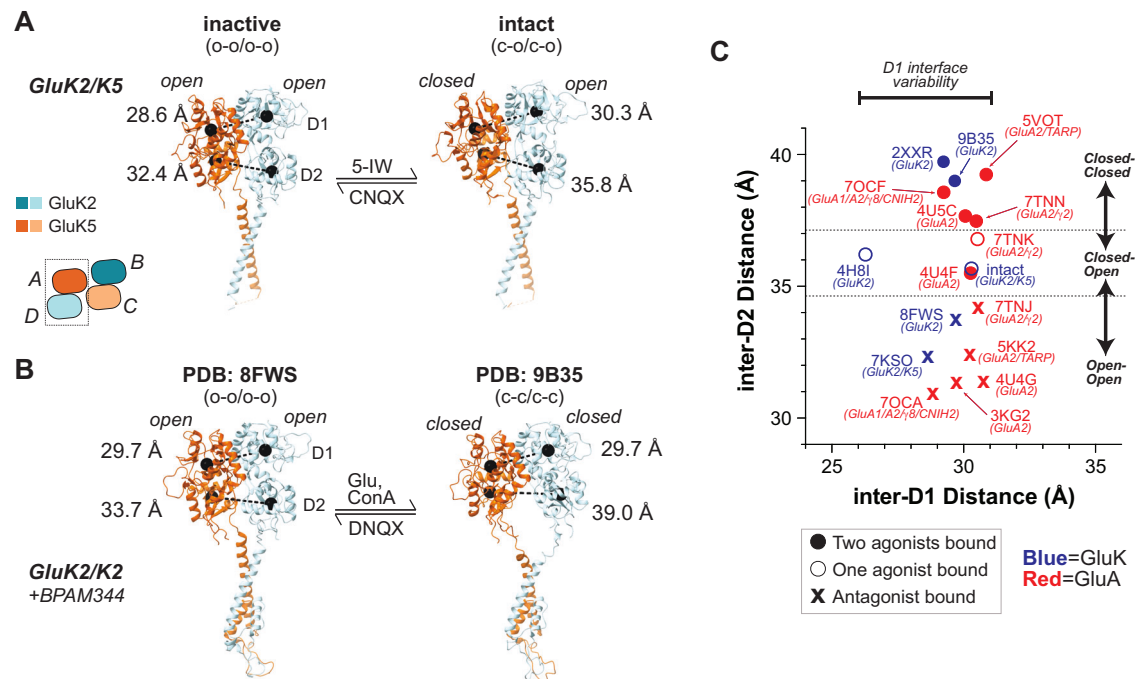


Fig. 2 | Intact LBD dimer arrangement depends on agonist occupancy.

A Structural comparison between antagonist-bound open/open and 5-IW-bound open/closed GluK2/GluK5 LBD heterodimers from inactive and intact structures. **B** Structural comparison between antagonist-bound open/open and glutamate-bound closed/closed GluK2 LBD homodimers in the presence of the desensitization blocker concanavalin A (ConA). Note: in these two structures, a positive allosteric

modulator (BPAM344) was also present. In **(A)** and **(B)**, center-of-mass distances between the upper (D1) and lower (D2) LBD lobes are shown. **C** Analysis of LBD dimer distances for antagonist-bound, partially occupied, and fully occupied AMPAR and KAR LBD dimers showing agonist occupancy-dependent lower lobe separation. Note: 4U4F has a partial agonist bound, and 4H8I is a LBD dimer-only structure.

(Fig. 2A; Supplementary Movie 1, 2). The center-of-mass distances between adjacent D1 and D2 lobes within the dimers increased by 1.7 and 3.4 Å, respectively, in the intact structure. The increase in D2 distances is similar to what was recently seen in a glutamate-bound GluK2 homotetramer structure in which desensitization was blocked by the lectin concanavalin A (ConA) and a PAM, BPAM433 and an open pore is observed⁴³. However, the extent of D2 loosening is less in the partially-liganded GluK2/GluK5 heteromer compared to these fully-liganded GluK2 homomers (Fig. 2B). A similar increase in D2 distances is associated with gating of GluA2 receptors (Fig. S5). Indeed, a broader analysis (Fig. 2C) revealed that both KAR and AMPAR subtypes show an agonist-dependent separation of D2, with closure of only one subunit having a partial effect, while variability is seen without a clear trend in the D1 distance. This analysis suggests that agonist-dependent separation of the D2 dimer interface is a common feature of iGluR gating, which scales with the degree of agonist occupancy such that two agonists are needed for maximal conformational changes.

Analysis of inter-dimer LBD conformational changes

In addition to LBD closure and dimer interface rearrangements, global conformational changes are seen in the LBD layer between antagonist-bound inactive and the intact, class 1 5-IW-bound structure. Figure 3A shows a rotation between dimers characterized by a $>10^\circ$ change in the angle between the helices G of the A/C (GluK5) subunits. This rotation is coupled to an overall expansion of the LBD layer, which results in an increase in the center-of-mass distances between all subunits (Fig. 3B; Supplementary Movie 3). Notably, the largest distance increase is seen between the GluK2 subunits in the B and D positions, while the GluK5 distance shows only a modest increase. These structural changes indicate that agonist occupancy of the GluK5 subunits in A and C positions is sufficient to drive global repositioning of the unoccupied GluK2 subunits in B and D positions. A similar, but larger, LBD layer global rearrangement is seen in the

BPAM433- and ConA-bound GluK2 homotetramer structures upon glutamate occupancy of all four subunits (Fig. S6A)⁴³. Further supporting agonist-induced large-scale LBD rearrangements, 3-D variability analysis (3DVA) of the intact GluK2/GluK5 cryo-EM map revealed a ‘rocking’ motion of the LBDs away from the midline, resulting in a decrease in compaction along the inter-dimer A-B and C-D interfaces (Fig. 3C; Supplementary Movie 4). 3DVA of the antagonist-bound GluK2/GluK5 map showed substantially less pronounced motions (Figure S6B; Supplementary Movie 5), consistent with our interpretation of inter-dimer motions as contributing to agonist-evoked receptor gating.

The above observations prompted us to closely analyze the intra- and inter-dimer LBD interfaces of the intact state structures (Fig. 4A). Overall, these interfaces are well conserved, and we thus focused on perturbations to interface residues that are unique to GluK4 and GluK5 relative to other KAR subtypes or KARs relative to AMPARs (Fig. 4B–D). We started by probing LBD dimer A–D (and B–C) interface residues (Fig. 4C), as the stability of this interface is known to be a key determinant of desensitization in KARs and AMPARs^{56–60}. One prominent site is GluK5 A504 in helix D, which is homologous to GluA2 L504, GluK2 Y521, and GluK3 H523. In AMPARs, this position strongly affects desensitization, as, for example, in GluA2(L504Y) desensitization is completely abolished^{56,61,62}. Interestingly, GluK1 and GluK2 naturally harbor a Tyr at this site (e.g., GluK2 Y521) (Fig. 4C), yet these receptors still desensitize as rapidly and as efficiently as AMPARs, indicating that distinct interfaces or dimer stabilities are at play for KARs. Other subtype differences are seen towards the edge of the GluK5–GluK2 dimer interface involving helices J and K, with a conserved Gln in KARs (GluK5 Q768) differing from a conserved Lys in AMPARs. In addition, N772 and N773 are unique to GluK5 and GluK4, while other KAR subunits contain at least one negatively charged residue at this position (Fig. 4C). Previous mutational studies were mostly performed in homomeric GluK2 receptors, where interface residues engage in

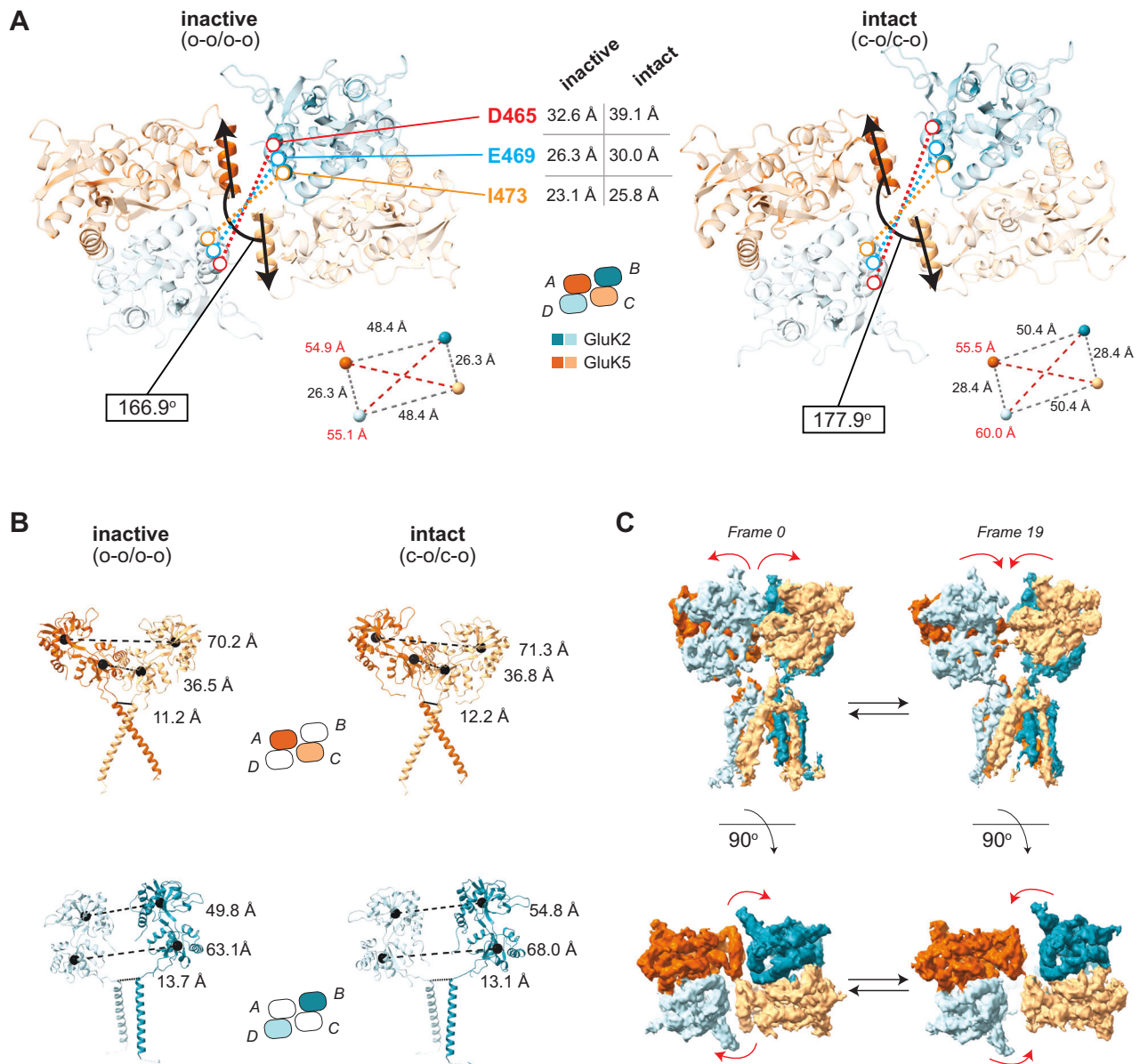


Fig. 3 | Activation-associated inter-dimer LBD conformational changes. **A** LBD layer (top view) of inactive and intact GluK2/GluK5 structures showing large-scale intersubunit rearrangement. Reported distances are between C alpha of chain B and D residues and angle is for helix G (residues 684–695) between chain A and chain C. Bottom insets show center-of-mass distances between LBDs, with a major increase in B-D distances. **B** Inter-dimer rearrangements between GluK5 subunits

(top) and GluK2 subunits (bottom). Center-of-mass distances are shown for D1 and D2, as well as the distance between the top of the M3 helix (Met648 for chains A/C; Val661 for chains B/D). **C** 3-D variability analysis shows inter-dimer motions in the intact state. Shown are two snapshots from the side (top) and top (bottom) for the major motion component detected. See also Supplementary Movies 3 and 4.

different types of interactions, depending on whether they are located in the A/C or B/D subunits. Restricting substitutions just to the GluK5 subunits (i.e., the A/C subunits in GluK2/GluK5 heteromers), provides a simpler-to-interpret scenario than introducing the same substitutions in homomeric receptors, where they engage in a second, separate set of interactions via the B/D subunits. Upon co-expression with GluK2, all GluK5 variants produced desensitizing glutamate responses and resulted in similar surface expression levels compared to GluK5(wt) (Fig. S7).

We first used long glutamate pulses (100 ms) to measure glutamate-induced desensitization of GluK2/GluK5 heteromers, which typically follow single exponential kinetics (τ_{desen} ; Figs. 4E, F; S7, and Table S2). However, when the same cells are exposed to brief (0.6–1 ms) pulses of glutamate, a slow deactivation becomes apparent, which is a unique feature of GluK2/GluK5^{18,23} and likely other GluK5-containing

heteromers²⁴. In these measurements only a fraction of receptors desensitize, whereas the 2nd decay phase can be attributed to the slow deactivation of GluK2/GluK5 heteromers, which reflects τ_{deac} (Fig. 4E, F). A second brief glutamate pulse gives a somewhat lower amplitude due to the desensitized receptor fraction, which can be used to estimate heteromer content in the recorded cell (see *Methods*). For GluK2/GluK5(wt) we found $\tau_{\text{desen}} = 3.3 \pm 0.1$ ms and $\tau_{\text{deac}} = 73.7 \pm 4.9$ ms (mean $\pm$ s.e.m, $n = 18$ cells; Fig. 4G).

Heteromers incorporating the GluK5(Q768K) variant in the A/C subunits opposing GluK2 Y521 in the B/D subunits showed only slightly accelerated desensitization kinetics compared to GluK2/GluK5(wt) (Figs. 4G; S7B, C). In contrast, the GluK5(A504Y) exchange, a substitution that is homologous to GluA2(L504Y), had no effect on desensitization, but showed a minor slowing of deactivation (Fig. 4G; S7B, C), which supports prior work indicating that the LBD

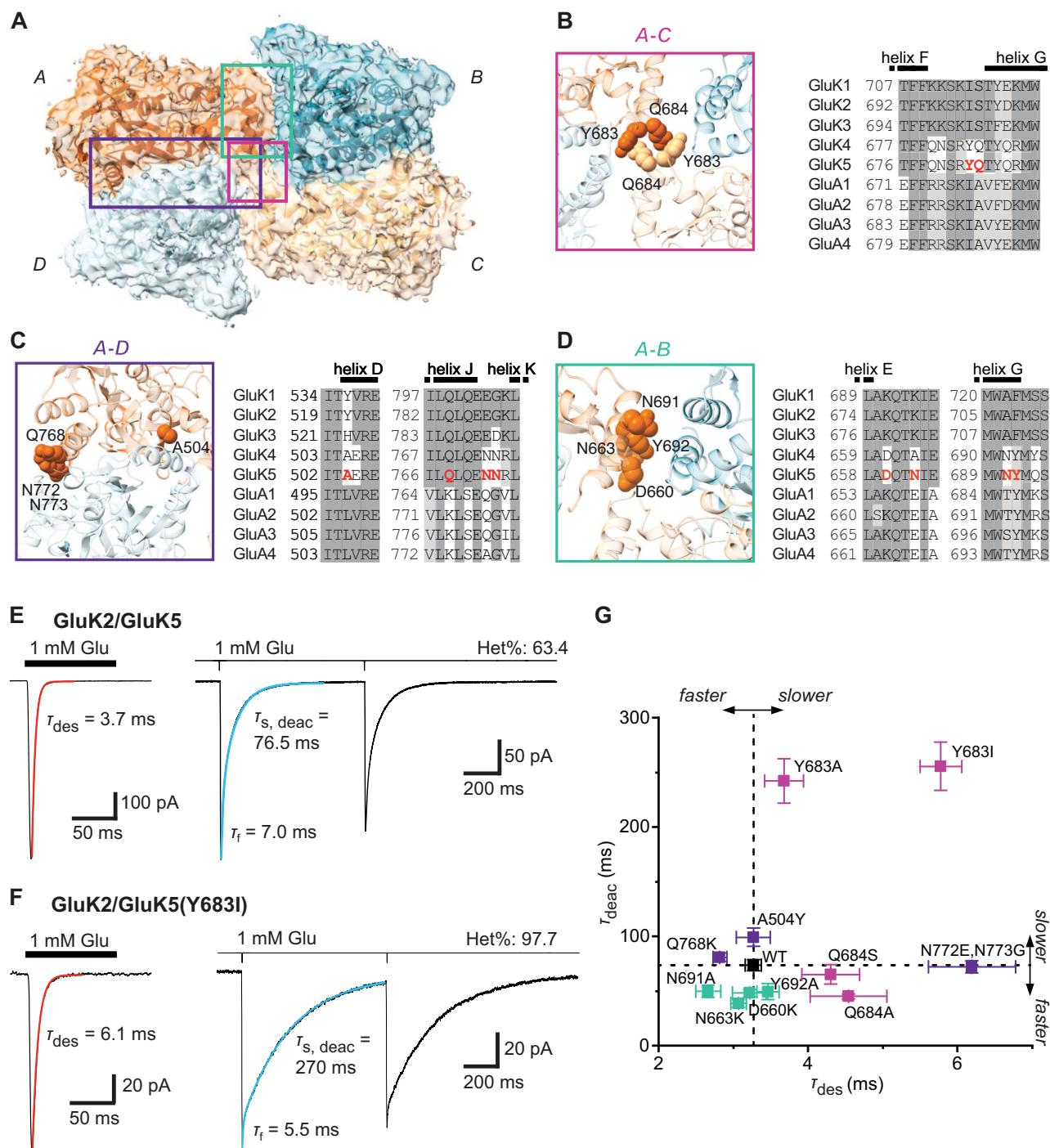


Fig. 4 | Intra- and inter-dimer interactions control GluK2/GluK5 deactivation and desensitization. A Subunit interfaces in the LBD layer (top view). Cryo-EM map displayed at a threshold of 0.132 overlaid onto the model with colored boxes defining the A-D dimer interface (purple), the A-C central cluster (magenta), and the A-B inter-dimer interface (turquoise). Side chain depictions (spheres) and local sequence alignments of the A-C inter-dimer central cluster (**B**), the A-D intra-dimer interface (**C**), and the A-B inter-dimer interface (**D**) focusing on residues that are

unique to GluK5 (substituted residues in red). **E, F** GluK2/GluK5(wt) and GluK2/GluK5(Y683I) currents in response to long (100 ms) glutamate pulses (left), showing desensitization, or two short pulses (0.6 ms) showing deactivation. Desensitization was fitted with a single exponential (red lines) and deactivation was fitted with a double exponential (blue lines). **G** Deactivation and desensitization time constants for all GluK2/GluK5 interface variants tested. Error bars show s.e.m. For details see Fig. S7 and Table S2.

dimer interface contributions may be different in KARs compared to AMPARs^{62–65}. However, the double substitution GluK5 (N772E, N773G), which mimics the residues present in GluK2, resulted in 1.8-fold slower desensitization (Figs. 4G; S7B, C), revealing a subtype-specific contribution to KAR desensitization.

As previously noted²³, GluK2/GluK5 heteromers show a more tightly packed LBD layer compared to GluK2 homomers and AMPARs.

Strikingly, in both inactive and intact structures, the two GluK5 subunits are in close proximity to each other, enabling the formation of a central cluster across dimers (Fig. 4A, B). In 3DVA this inter-GluK5 interaction is maintained during the aforementioned LBD layer conformational changes (Fig. 3C), suggesting that the tip of helix G of subunits A/C (GluK5) may serve as a hinge region. Inspection of the electron densities around this region revealed a dense cluster in the

center of the four LBDs primarily formed by residues Y683 and Q684 from the two GluK5 subunits (Fig. 4B). Y683 and Q684 are unique side chains found in the ‘high-affinity’ GluK4 and GluK5 subunits, whereas the region is highly conserved in all other AMPAR and KAR subunits with Ile and Ala/Ser residues at these two positions, respectively (Fig. 4B).

We thus tested the functional effects of mutation to these residues by substituting Y683 either to Ala or Ile, the residue present in GluK2. Both Y683I and Y683A substitutions had considerable effects on gating kinetics (Figs. 4F, G; S7B, C). Desensitization of GluK2/GluK5(Y683I) heteromers was 1.8-fold slower compared to GluK2/GluK5(wt). A more pronounced effect was seen for the deactivation kinetics, which increased to 255.5 ± 22.2 ms in GluK2/GluK5(Y683I) (mean $\pm$ s.e.m). Also, deactivation of non-desensitizing 5-IW-induced currents was slowed 5.4-fold for this substitution (Fig. S7H, I). However, glutamate dose-response analysis of GluK5(Y683I) showed no major changes in apparent affinity (EC_{50}) compared to GluK5(wt), when measured in the presence of a binding-impaired GluK2 subunit GluK2(T690V) (Fig. S7F, G). The Y683A exchange resulted in a similar slowing of deactivation after glutamate and 5-IW application, but a less pronounced effect on desensitization (Figs. 4G; S7B, C, H, I). Substitutions of the other central cluster residue, Q684S and Q684A, had only minor effects on desensitization and deactivation (Figs. 4G; S7B, C). These data suggest that GluK5 residues at the tip of helix G of the A/C subunits are involved in cross-dimer interactions with the opposing GluK5 subunits and that this interaction plays a role in controlling gating.

Last, we probed residues in the A-B (and C-D) inter-dimer interfaces, which are formed by the D2 domains of GluK5 making contact with the GluK2 subunits right above the M3-S2 linkers. According to the intact-state 3DVA, the packing of these interfaces is rather dynamic, allowing for dense and loose packing during the identified LBD layer motions. Here we focused on the GluK5-specific residues D660 and N663, as well as N691 and Y692 (Fig. 4D). Substitution to the corresponding GluK2 residues (D660K, N663K, N691A and Y692A) led to faster deactivation kinetics, but had minimal effects on desensitization (Figs. 4G; S7A–C), suggesting that this interface may contribute to the uniquely slow deactivation kinetics of GluK5-containing receptors.

In summary, several GluK5-specific residues positioned in the LBD interfaces had substantial effects on the gating of GluK2/GluK5 heteromers (Fig. 4G). One of the substitutions in the A-D (B-C) interface had a clear effect on desensitization, which is in line with prior work on homomeric KARs and AMPARs that suggests that the stability of this interface is critical for controlling desensitization. More unexpectedly, inter-dimer interactions seem to be involved in bidirectionally controlling the uniquely slow deactivation of GluK5-containing receptors^{18,23,24} and modulating the rates of desensitization. Together, these data suggest that coordinated motions and intersubunit interactions across the entire LBD layer, rather than merely between LBD dimers, control GluK2/GluK5 gating.

Rearrangement of LBD-TMD linkers and pore expansion

The LBD layer rearrangements described above raise the question of how these conformational changes may be transmitted to the pore domain. The M1, M3, and M4 helices which we resolve in our structures are each connected to the LBD via linkers which connect to secondary structure elements within the upper (M4 linker) and lower (M1, M3 linkers) LBD lobes (Fig. 5A). Analysis of the pore diameter of the intact structure shows a similar constriction at 3 sites (T652/T636, A656/A640, and T660/T644 in GluK2/GluK5) as seen in inactive and fully ruptured structures (Fig. 5B), consistent with a closed ion channel. However, clear TMD helix repositioning is observed in the intact structure compared to the antagonist-bound state, showing an overall expansion of the transmembrane domain (Supplementary Movie 6;

Fig. S8). This transmembrane domain dilation, combined with the observed closed pore state, suggests that our intact structure represents a pre-active state. The lack of pore opening under these conditions is consistent with the low open probability of KARs⁶⁶ in the absence of PAMs and the partial activation observed with 5-IW (Fig. S1).

To gain insight into the conformational changes that may precede pore opening and enable LBD-TMD allosteric coupling, we analyzed pore linker and TMD helix repositioning in our intact pre-active structure relative to the antagonist-bound inactive GluK2/K5 structure (Fig. 5C–H). Upward motions of 1–3 Å relative to the TMD were observed for the M1, M3, and M4 helix linkers for both GluK2 (Fig. 5C–E) and GluK5 subunits (Fig. 5F–H). This repositioning is accompanied by a small increase in the separation between helices, which is largest (1.5–4 Å) at the cytoplasmic end of each helix. This leads to an overall expansion of the TMD (Supplementary Movie 6). The 2-fold symmetry of the LBD layer leads to the GluK2 (B/D) subunits being in pore-distal M3 linker positions (Fig. 5D) while GluK5 (A/C subunits) occupy pore-proximal positions (Fig. 5G). Consistent with prior structural studies of AMPARs^{48–51,53}, the M3 helix linkers in the B/D position also undergo an outward motion (Fig. 5D), which, together with the aforementioned upward motions, is likely to increase tension and to allow mechanical pulling of the M3 helix to trigger pore opening. A similar outward motion is seen for the M1 (Fig. 5C) and M4 linkers (Fig. 5E) in the B/D positions. It is notable that these gating-associated conformational changes occur despite the GluK2 LBDs remaining open, revealing that ligand binding to only the GluK5 subunits (A/C subunits) is sufficient to drive global gating-associated conformational changes that are propagated to the TMD. In contrast to the linker separation observed for the B/D positions, the A/C linkers on GluK5 subunits undergo a modest tightening in the intact structure (Fig. 5F–H). This differs from the fully occupied, desensitization-blocked open-state GluK2 structure⁴³, where the M3 linkers undergo upward and outward motions at all four positions, which are coupled to an M3 helix kink, which is not seen in our pre-active structures.

Structural features of partially and fully ruptured LBD states

As described above, partial 5-IW agonist occupancy also led to the formation of a partially ruptured state (class 2; Fig. 1C–E) with one closed/open LBD dimer being intact and the other ruptured. The population of this partially ruptured state in the presence of non-desensitizing ligand conditions suggests that it reflects an intermediate which, owing to the fact that it has not yet undergone full rupture and formation of a desensitization-ring^{23,40}, may readily interconvert with the intact state. Importantly, the partially ruptured state shows a closed pore conformation that is comparable to intact and fully ruptured states (Fig. S9). We examined the intact A-D LBD dimer of the partially ruptured structure and noted that, while the general orientation is similar to that seen in the fully intact state, a reshaping of the dimer interface occurs, which includes tightening along the D2-D2 surface and a $\sim 5^\circ$ inter-subunit rotation (Fig. 6A; Supplementary Movie 7, 8). This subtle conformational rearrangement suggests a potential stabilization of the inter-subunit interaction that may prevent rupture of the A-D LBD dimer and subsequent functional desensitization. In addition, the ruptured B-C dimer shows distinct inter-subunit interface angles in the partially versus fully-ruptured states (Fig. 6B), indicating that the two LBD dimers do not change their conformation in a fully independent manner. In addition to changes in the LBD dimer organization, partial and full rupture alter the LBD inter-dimer interfaces (Fig. 6C, D; Supplementary Movie 9, 10). In the partially ruptured states, the central cluster remains partially formed with GluK5 subunits in close proximity to each other, although limited resolution prevents us from confidently modeling side chain positions. However, LBD layer reorganization leads to Y683 repositioning such that it may still interact with its ruptured dimer partner. An inter-dimer

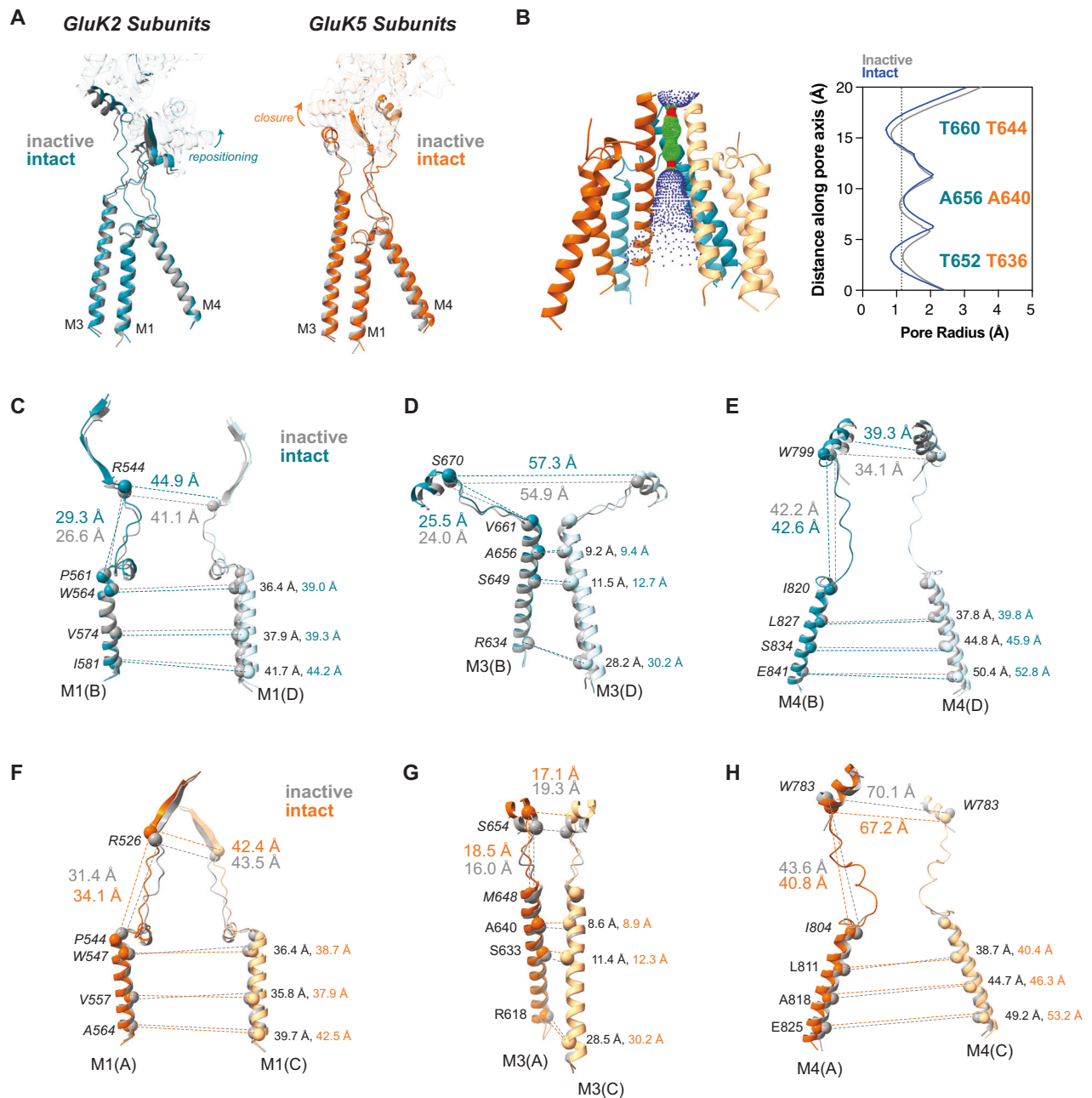


Fig. 5 | Agonist-evoked rearrangements of pore linkers and TMD helices.

A Structural alignment of TMD and lower LBD portion for a single GluK2 (left) or GluK5 (right) subunit in inactive (grey) or intact (turquoise or orange) states, showing subtle repositioning of helices and linkers. **B** Left, visualization of the pore volume in the channel of the intact structure. Regions with pore radius less than

1.15 Å are shown in red. One receptor subunit is hidden for clarity. Right, pore profiles for GluK2/GluK5 in inactive and intact state with chain D hidden for clarity. Vertical dashed line marks 1.15 Å pore radius. Structural alignments of M1 (**C**), M3 (**D**), and M4 (**E**) helices showing inter-helix separation and linker repositioning. **F–H**, Same as C–E for GluK5 subunits.

A-B interface is maintained but repositioned in the partially ruptured state such that an entirely different surface of GluK2, consisting of helix G and the helix F-G linker, is positioned opposite of GluK5 to form an inter-dimer interface. In contrast, the C-D interface is fully lost. In the fully ruptured state, the central cluster is replaced by the formation of a desensitization ring, and both Y683 and Q684 are positioned closer to their adjacent GluK2 subunits than to the opposing GluK5 subunit. In addition, distinct A-B and C-D interfaces are formed in the fully ruptured state compared to the intact state. These altered inter-subunit interfaces may contribute to the complex effects on deactivation and desensitization seen upon substitution of the high-lighted residues (Fig. 4G).

Both partial and full LBD rupture are associated with linker repositioning and pore domain motions. Overall, partial rupture is accompanied by a modest TMD contraction relative to the intact state (Supplementary Movie 11), while full rupture leads to a more dramatic TMD contraction relative to the partially ruptured state (Supplementary Movie 12). As seen before for KARs^{23,40,43}, LBD dimer rupture is associated with repositioning of the GluK2 subunit M3-S2 linker in B and D positions (Fig. 6E). In the GluK5 A and C positions, the M3 linker shows more modest rearrangements (Fig. 6F). Interestingly, the linker position for the intact subunit A of the partially ruptured structure remains ~3 Å elevated relative to the fully ruptured state, implying that some linker tension is maintained in this state. The S2-M4 linker helix

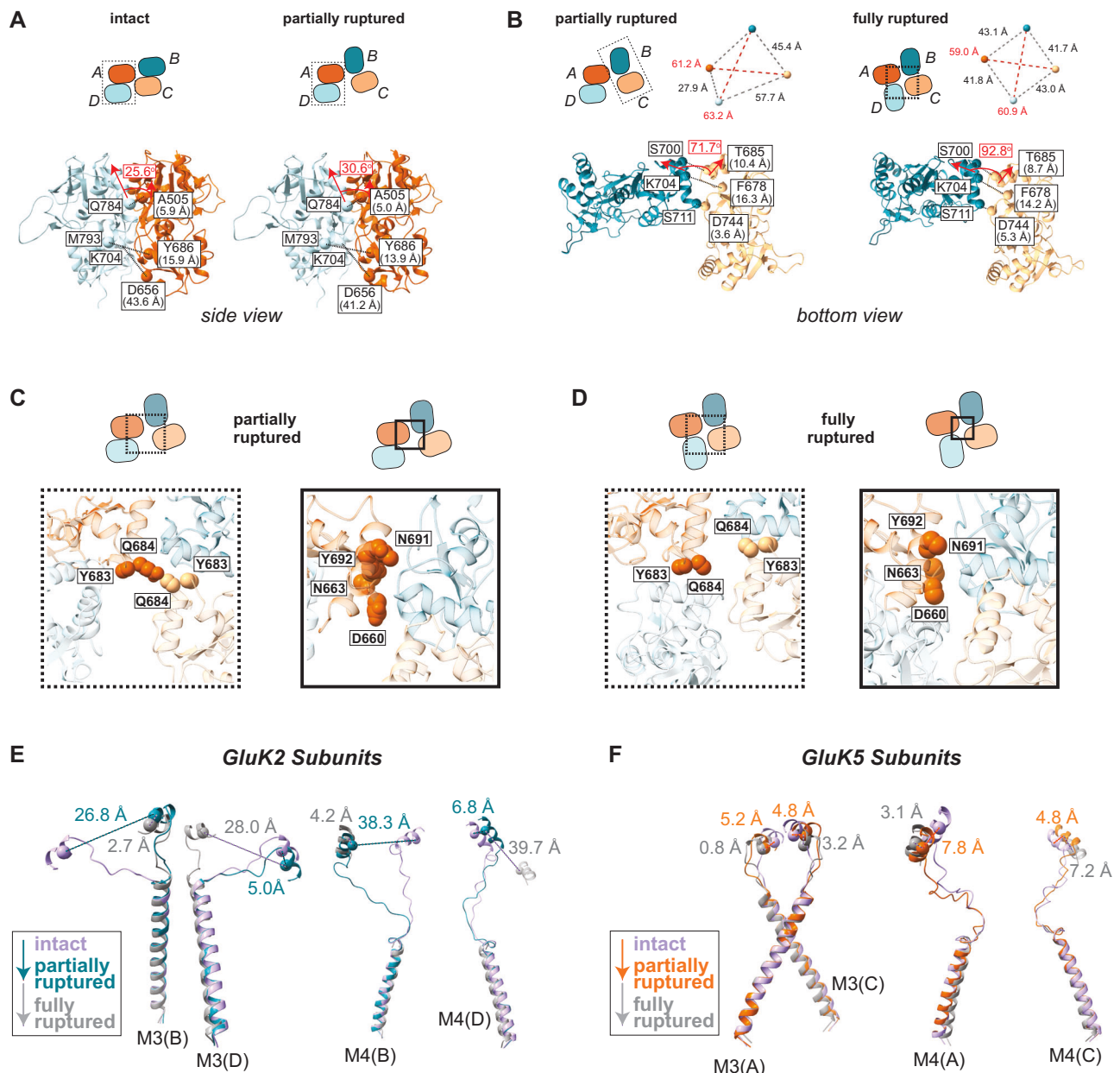


Fig. 6 | Conformational rearrangements associated with LBD dimer rupture. **A** Distinct A-D LBD heterodimer arrangements for intact (left) versus partially ruptured (right) GluK2/GluK5 structures. The angle (red) shown is that between helix E of chain A (aa504-507) and chain D (aa774-787). **B** Distinct LBD tetramer arrangements (top) and B-C LBD interfaces (bottom) for partially ruptured (left) and fully ruptured (right) states. All distances shown are for Calphas and the angle

shown is that between aa671-676 (chain B; helix E) and aa655-660 (chain C). Structural depictions highlighting key interfacial residues of GluK5 across agonist-bound partially (**C**) and fully ruptured (**D**) states. Structural alignments of M3 (left) and M4 (right) helices showing linker repositioning upon partial and full LBD dimer rupture in GluK2 (**E**) and GluK5 (**F**) subunits. See also **Supplementary Movies 7–12**.

of the B and D subunits also shows dramatic repositioning upon LBD rupture (Fig. 6E), while more subtle changes are seen in the A and C subunits with distinct positions for intact, partially ruptured, and fully ruptured structures (Fig. 6F). These observations are consistent with stepwise, concerted changes that modify the coupling between the LBD and TMD layers upon partial and full LBD rupture.

Correlating activation and desensitization with LBD layer conformational changes

Motivated by the range of LBD layer conformational changes observed upon partial agonist occupancy of GluK2/GluK5, we asked if similar motions would be observed with another GluK5-selective agonist, AMPA⁵⁵. In functional patch-clamp measurements we observed no

response to AMPA for GluK2 homomers and clear responses for GluK2/GluK5 heteromers (Figs. 7A, B; S10). However, unlike what was seen with 5-IW (Fig. 1A), a slow (~60 ms), yet clear, ~30% desensitization was observed upon application of AMPA (Figs. 7B; S10). Recovery from this partial desensitization occurred on the 40 ms time scale (Fig. S10). We speculated that AMPA may show partial agonism at GluK2 subunits in the heteromer context, as seen previously in both functional and single-molecule FRET analysis of LBD closure^{67,68}. Indeed, AMPA-induced GluK2/GluK5 desensitization was suppressed in the presence of the GluK2-specific antagonist kynurenic acid (Figs. 7C; S10K-N), indicating that AMPA also acts through the GluK2 subunits in GluK2/GluK5 heteromers and revealing homo-tetramer versus hetero-tetramer context-dependent changes in GluK2 pharmacology.

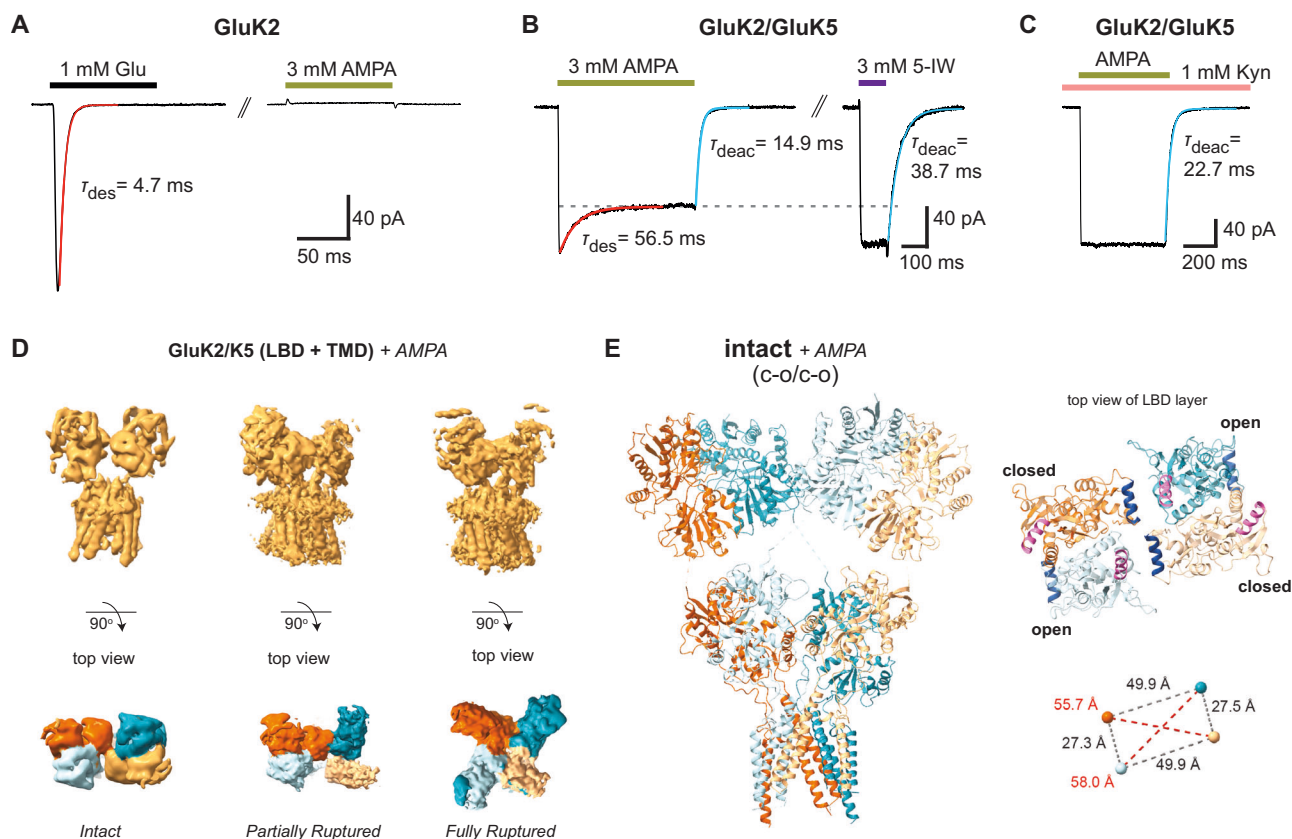


Fig. 7 | AMPA-driven GluK2/GluK5 desensitization and structural rearrangements. **A** Glutamate but not AMPA produces large inward currents in GluK2-expressing cells. **B** In GluK2/GluK5-expressing cells, AMPA produces large inward currents which undergo partial desensitization in contrast to what is seen with 5-IW (see also Fig. S10). **C** AMPA-induced desensitization of GluK2/GluK5 was blocked by

a GluK2-selective antagonist (kynurenic acid). **D** Maps of distinct LBD-TMD reconstruction classes for GluK2/GluK5 showing intact, partially, and fully-ruptured LBD states in the presence of AMPA. **E** Model for the intact AMPA-bound GluK2/GluK5 structure showing a similar conformation to the intact 5-IW-bound model with closed-open LBDs and associated LBD arrangement.

We then asked which conformations are stabilized by AMPA, as this may provide insight into the structural basis of KAR desensitization. Using a similar procedure to that was used for our 5-IW structures, we determined cryo-EM structures of GluK2/GluK5 heteromers after brief incubation with AMPA, followed by sample vitrification. Compared to our 5-IW data collection, a wider range of classes was observed, including lower resolution classes with varying degrees of LBD dimer rupture. We initially resolved an intact LBD state (189,632 particles) and two partially ruptured LBD states (215,800 and 175,570 particles) (Fig. S11A–C. 3DVA of the combined partially ruptured state classes revealed a wide range of LBD motions, including apparent full rupture of both LBD dimers (Supplementary Movie 13). Further processing of the combined LBD-TMD module of ruptured states revealed a variety of medium-resolution states. This includes a class with apparent rupture of both LBD dimers representing ~25% (98,296/391,340 = 25.1%) of the particles containing non-intact LBD interfaces (Figs. 7D; S11D). Only the intact class produced a high-resolution state which showed closed GluK5 LBDs and open GluK2 LBDs (Fig. S12A) and similar LBD layer and pore linker conformational changes relative to inactive GluK2/GluK5 as those observed with 5-IW (Fig. 7E; S12B–E). Together these data point to a primary population of intact and partially ruptured states along with a subpopulation of fully ruptured conformations, in the presence of AMPA.

Discussion

Our structural and functional data collected under distinct orthosteric ligand conditions in the absence of desensitization blockers or PAMs provide insight into the mechanisms of gating for heteromeric GluK2/GluK5 tetramers, a highly prevalent kainate receptor in the mammalian

brain. Adding to the plethora of prior structures, our structures showing intact and partially-ruptured states under non-desensitizing ligand conditions (i.e. 5-IW) and fully-ruptured states only under desensitizing ligand conditions (e.g. AMPA; glutamate), allow us to propose a working model of KAR gating (Fig. 8). As previously proposed based on kinetic analysis of functional data^{20,31,32}, our model posits the presence of one pre-active state (P) and two desensitized state configurations, which we correlate with partially (D) and fully ruptured (D') states, respectively. In contrast to prior work, we posit that the D' states are only accessible to receptors with at least 3 agonists bound, consistent with the lack of full LBD rupture seen with 5-IW. Notably, structures have been determined across the range of proposed states, with partial rupture observed in the absence of agonist^{41,42,45}, with 2 agonists (this study), with 3/4 agonists in the absence of desensitization blockers (this study), 4 partial agonists^{43,45}, or 4 agonists in the presence of PAM^{43,45} or LBD interface mutants⁴⁷. In contrast, full rupture has only been observed with at least 3 agonists bound (this study; refs. 10,23,40–43,45,47). It is worth noting that our ligand occupancy analysis and thus our model is based on sequential occupancy of the A/C positions followed by the B/D positions and it is possible that full occupancy of one of the two LBD dimers may be sufficient to stabilize D' and promote desensitization. Interestingly, Yelshanskaya et al.³⁰ used sub-saturating glutamate concentrations and rapid freezing to capture partially occupied states of homomeric GluA2 receptors via cryo-EM and found evidence that the A/C positions are preferentially occupied prior to the B/D positions, suggesting that even in homodimers, ligand binding may be biased. In order for occupancy of the D2 state not to drive functional (i.e., lasting) desensitization, it is required that rapid exchange can occur between this

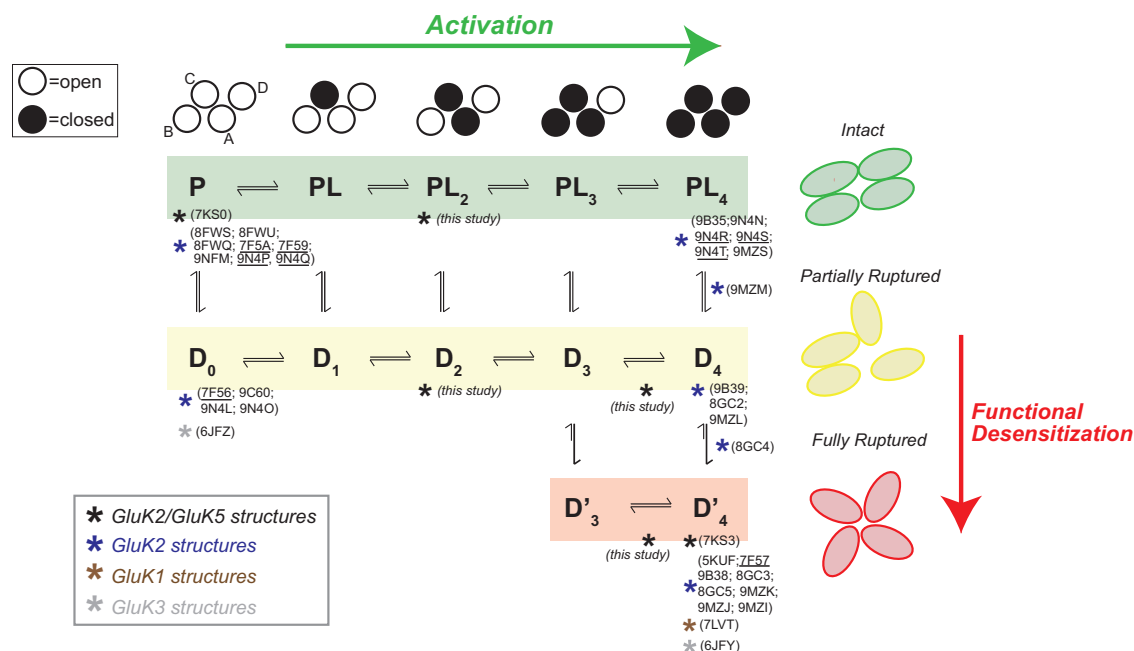


Fig. 8 | A working model of ligand occupancy-dependent KAR gating. Simplified state model showing differential agonist occupancy-dependent activation and desensitization. Ligand binding to at least three subunits is required to access the fully ruptured D' state, which is associated with long-lasting functional desensitization. PDB numbers for reported KAR structures are listed for each state, with those obtained in the presence of the Neto2 accessory protein underlined. Note

that some P and D₀ state structures were obtained with antagonists and that AMPA-bound partially- and fully-ruptured states have insufficient resolution to determine if 3 or 4 LBDs are closed and are thus placed between D₃ and D₄ and D'₃ and D'₄. 8GC4 may represent an intermediate between partial and fully ruptured states and 9MZM may represent an intermediate between intact and partially ruptured states. 9MZx structures are of a cross-linked GluK2 mutant (K676C/N802C).

partially ruptured state and intact states. In line with this, two recent studies reported partially ruptured D₀ states for apo GluK2^{45,46}, which, based on functional data, would have to be inactive but not desensitized. In contrast, the functional desensitization seen upon >2 agonists binding and the much slower time scale of recovery from desensitization imply a slow interconversion from fully ruptured D' states back to partially ruptured and intact states. Importantly, this highly simplified model is not meant to fully describe KAR gating, but rather to serve as a working guide for future studies. This model does not take into account the asymmetric nature of the subunits, which gives rise to a total of ten different occupancy configurations or the many possible further structural and functional intermediates, including different open states, and does not address cooperativity between subunits. Future structural, conformational, and functional work, ideally including direct measurements of relevant rate constants and defined occupancy patterns in combination with detailed theoretical modeling, are needed to produce a more integrated view of ligand occupancy, conformational rearrangements, and conductance.

Our GluK2/GluK5 structures revealed a particularly tight LBD layer packing which appears to facilitate inter-dimer interactions and may contribute to the subtype-specific properties of different iGluRs. Substitutions to residues in either a central cluster formed between GluK5 subunits in the A-C position or to residues in the A-B and C-D interface had effects on both deactivation and desensitization. GluK2/GluK5 heteromers provide a unique test bed for such studies, since substitution in GluK5 subunits allows for defined changes only in the A/C positions without affecting the distinct interactions formed by the same residues in the B/D positions. Indeed, the deactivation rate constants of GluK2/GluK5 heteromers were dramatically slowed into the 250 ms range when performing GluK5 central cluster substitutions at the tip of helix G, i.e. GluK5(Y683A) and GluK5(Y683I) (Figs. 4F,G; S7). In contrast, several substitutions in the A-B (and C-D) interfaces caused acceleration of deactivation, albeit not reaching the speed of GluK2, which deactivates within a few milliseconds. This

provides a starting point for further studies addressing the slow deactivation mechanism that is unique to GluK5-, and possibly GluK4-, containing receptors and which is likely to have important physiological consequences^{18,23,24}. However, while our results provide strong evidence that inter-subunit interactions beyond the canonical dimer interface are important for KAR function, the complex re-shaping of these interfaces across intact, partially-ruptured, and fully-ruptured states (Fig. 6C, D) challenges interpretation.

Overall, inter-dimer iGluR interfaces have received less attention compared to the intra-dimer interface. However, multiple studies have shown that A-C and A-B (and C-D) residues can be crosslinked across the LBD layer of GluA2, GluK2, or NMDARs and that this can alter current amplitude, desensitization, and recovery from desensitization in a site- and subunit-specific manner^{69–76}. In a notable study, A-C intersubunit crosslinking in GluA2 homotetramers was found to occur in a state-dependent manner and to trap the receptor in a partially activated and apparently partially liganded state⁷¹, potentially similar to the partially liganded pre-active intact state we observe for GluK2/GluK5 in this study. Further A-C crosslinking studies showed that the presence of auxiliary subunits can maintain GluA2 in a compact state without LBD layer opening⁷⁶. Similarly, structure-guided simulations identified a rolling motion between LBD dimers in a NMDAR, which controls both gating of the channel and allosteric modulation via the ATD⁷⁴. In addition, helix G mutations to the desensitization ring of GluK2 accelerated recovery from desensitization⁴⁰. Our work and these studies together motivate future structural, spectroscopic, and computational work to dissect this complex network of intra- and inter-dimer interactions that provide subtype-specific control over gating kinetics and may potentially serve as additional sites for subtype-specific allosteric drugs. Here, the intact state provided us a side-chain model of the GluK5 LBD, which can serve as a basis for computational and functional investigation of ion binding and allosteric modulator binding, which both occur at the LBD intra-dimer interface and have so far been studied primarily in the context of

homomeric KARs^{77–82}. With respect to the LBD interfaces, it is worth noting that GluK5 and GluK4 do not differ in critical positions (see also Fig. 4B–D), suggesting that similar intra- and inter-dimer interactions may be involved in regulating GluK4-containing heteromers. In contrast, differences are observed across other GluK and GluA subtypes, raising the likelihood that the central cluster and other inter-dimer interactions identified here may be differentially tuned across iGluRs.

Our AMPA-bound structures (Fig. 7D) and many prior KAR structures collected under desensitizing conditions show large-scale LBD dimer rupture and formation of a desensitization ring^{10,23,40–43}, even in the presence of Neto2⁴², an accessory subunit that slows desensitization. Given that structural states cannot readily be assigned to functional states and that AMPAR structures do not typically show a similar conformational change, the question arises whether this rupture represents an artifact of the in vitro, purified cryo-EM conditions or a truly desensitized state^{43,83}. Here we find partially ruptured states for a non-desensitizing partial agonist occupancy with 5-IW, while full rupture was limited to AMPA, which causes some functional desensitization (Fig. 7B). Other recent work has shown that domoate, a partially desensitizing agonist, induces a distinct desensitization ring conformation⁴⁵. Moreover, crosslinking of the A-B and C-D inter-dimer LBD interface was recently shown to prevent rupture by stabilizing a shallow-desensitized conformation, but this state likely remains conductive⁴⁷, further supporting the need for full dimer rupture to drive functional desensitization. Notably, KARs show stable ATDs very conformationally (Figure S3) with minimal interactions with the LBDs, which appears to enable major, yet reversible LBD layer changes. In addition, KARs show slow and complex recovery-from-desensitization kinetics^{19,20} which is consistent with larger structural changes associated with desensitization and may be related to the slower neuromodulatory roles of KARs compared to AMPARs^{1,4,5}. Intriguingly, GluN1/GluN3A glycine-gated NMDARs were recently shown to undergo a similar large-scale LBD dimer rupture⁸⁴, consistent with the slow (>2 s) recovery from desensitization seen for this unique subtype⁸⁵ and suggesting a related mechanism to KARs. Furthermore, there is evidence that in AMPARs, desensitization-related LBD layer changes can be larger than previously thought, as partial LBD dimer rupture has now been reported for GluA1/γ3 and a variant GluA2(F231A)/γ2, probably resulting from destabilized ATD configurations⁸⁶. Importantly, the LBD dimer rupture is not as complete in these structures as in KARs, but it does result in a substantial slowing of recovery from desensitization.

Our data also sheds light on the activation of partially agonist-occupied iGluRs. We find that a single bound agonist per LBD dimer is sufficient to produce substantial currents, which are associated with a plethora of conformational changes, including LBD closure, D2 domain separation, inter-dimer LBD layer motions, LBD-TMD linker repositioning, and pore expansion. Our analyses suggest that many of these conformational changes may be graded with ligand occupancy, as fully liganded intact structures typically show similar, but larger conformational changes^{30,43,48,50–52}. Most notably, our data show that A/C subunit occupancy is sufficient to drive the B/D M3-S2 linker separation that has long been hypothesized to be a key step in activation. Interestingly, this gating-dominance of the B/D subunits, in terms of their tighter coupling to the M3 helix and their larger scale structural rearrangements, is also exerted when only the A/C subunits are occupied. However, it remains unclear if there are differences in the extent of activation that occurs with liganding of A/C versus B/D subunits.

Overall, our study reveals distinct structural features of partially occupied KARs and the relationship between ligand occupancy, LBD dimer rupture, LBD layer rearrangements, and desensitization. This motivates further work on the distinct structural, functional, and context-dependent pharmacological properties of heteromeric KARs, as our findings bolster prior evidence that heteromerization is a key

physiological feature of this neuromodulatory receptor class. Future work can extend the approaches outlined here to probe the structure and function of the diversity of KAR di-, tri-, and tetra-heteromers and their association with auxiliary subunits and other regulatory proteins.

Methods

Construct design, protein expression, and purification for Cryo-EM

GluK2 and GluK5 constructs were previously designed and validated and referred to as GluK2em and GluK5em, with the heteromer described as GluK2/K5em²³. Briefly, GluK2 (Uniport: P42260) and GluK5 (Uniport: Q63273) constructs were designed from full-length rat genes and cysteine mutations (C576V and C595S in GluK2; C559V, C578S, C619I, and C813A in GluK5) were incorporated, the C-terminal region of GluK5 after the M4 helix was truncated after residue 829, and the GluA2 ‘tail’ (YKSRAEAKRMK) was added to the C-terminal of GluK5 to promote subunit expression and heteromer monodispersity. In addition, the GluK2 subunit was RNA-edited at position 567 (I to V). A Twin-Strep tag was fused to the C-terminus of GluK2, while EGFP and a 1D4 tag were added to the C-terminus of GluK5. The heteromeric GluK2/K5em receptor was expressed in HEK293S GnTI[−] cells using the BacMam system, solubilized in 100 mM dodecyl-β-D-maltoside (DDM, Anatrace), 0.5% cholesterol hemisuccinate (CHS, Anatrace), followed by sequential Twin-Strep (GluK2) and 1D4 (GluK5) affinity purification and Superose 6 size-exclusion chromatography²³.

Cryo-EM sample preparation and data acquisition

Cryo-EM samples were prepared using purified GluK2/K5em receptor (4.7 mg/mL). UltrAuFoil 1.2/1.3 300 mesh grids (Quantifoil) were plasma-treated and rendered hydrophilic by reaction with PEG-thiol⁸⁷. For GluK2/K5em-5-IW-Kyn, kynurenic acid (sodium salt, Tocris) was added to the GluK2/K5em receptor in a final concentration of 1 mM and incubated for 10 min. 2.2 μL of the GluK2/K5em receptor incubated with kynurenic acid was added to the grid. 0.4 μL of 2 mM 5-iodowillardiine (Tocris) was added to the protein in grid, mixed quickly, and immediately blotted (2 s blot time, −2 or −4 blot force) and plunge-frozen in liquid ethane using a Vitrobot Mk IV (Thermo Fisher). For GluK2/K5em-AMPA, 2.2 μL of the GluK2/K5em receptor was added to the grid. 0.4 μL of 6.5 mM AMPA (Tocris) was added to the protein in grid, mixed quickly, and immediately blotted (2 s blot time, −2 or −4 blot force) and plunge-frozen in liquid ethane using a Vitrobot Mk IV (Thermo Fisher).

Single particle images of GluK2/K5em-5-IW-Kyn were collected with Legicon⁸⁸ using a Titan Krios electron microscope (Thermo Fisher) operated at 300 kV and a nominal magnification of 105,000X and equipped with a GIF (Gatan) and K3 camera (Gatan) set in high-resolution mode (0.426 pixel size). Movies were collected at a dose rate of 21.29 e[−]/Å²/s with a total exposure of 2.40 s, for an accumulated dose of 49.56 e[−]/Å². Intermediate frames were recorded every 0.05 s for a total of 48 frames per micrograph. A total of 8068 images were collected at a nominal defocus range of 1.3–2.0 μm. Single particle images of GluK2/K5em-AMPA were collected with Legicon using a Titan Krios electron microscope (Thermo Fisher) operated at 300 kV and a nominal magnification of 81,000 and equipped with a GIF (Gatan) and K3 camera (Gatan) set in counted mode (1.083 Å pixel size). Exposures were made with an average defocus value of −1.6 μm, dose fractionation into 40 frames, a total exposure time of 2 s, and a total dose of 51.28 e[−]/Å². A total of 13,696 movies were recorded.

Cryo-EM image processing

Movie stacks were corrected for beam-induced motion (with twofold binning for super-resolution K3 data) in Relion 3.1⁸⁹ yielding dose-weighted images with an image pixel size of 0.852 Å for the GluK2/K5em-5-IW-Kyn dataset, and 1.083 Å for the GluK2/K5em-AMPA dataset. These images were used for contrast transfer function (CTF)

estimation with CTFFIND4.1⁹⁰. Reference-free particle auto-picking (Laplacian-of-Gaussian) followed by 2D-template-based picking was done in Relion. The auto-picked particles were extracted with a box size of 416 pixels (GluK2/K5em-AMPA) or 480 pixels (GluK2/K5em-5-IW-Kyn), binned to a box size of 128 pixels, and imported into cryoSPARC⁹¹. One round of 2D classification was performed for both the GluK2/K5em-5-IW-Kyn and GluK2/K5em-AMPA datasets. Ab initio 3D reconstruction and several rounds of heterogeneous refinement with C1-symmetry were used for both datasets. This process removed false positives from auto-picking and isolated interpretable particle sets.

The resulting particles were re-extracted with a box size of 416 pixels (GluK2/K5em-AMPA) or 480 pixels (GluK2/K5em-5-IW-Kyn) without binning and imported into cryoSPARC. Particles were re-centered during re-extraction. Ab initio 3D reconstruction and several rounds of heterogeneous refinements were again performed with C1-symmetry to obtain the final set of particles. The final 3D reconstructions were obtained by non-uniform refinement in cryoSPARC. For the GluK2/K5em-5-IW-Kyn dataset, there were two classes with intact and partially ruptured LBD conformations. The global resolutions were calculated using the FSC with a threshold of 0.143 as 3.83 Å (GluK2/K5em-5-IW-Kyn – intact LBD) and 3.52 Å (GluK2/K5em-5-IW-Kyn – partially ruptured LBD). For the GluK2/K5em-AMPA dataset, there were three classes with one intact and two partially ruptured LBD conformations. The global resolutions were calculated using the FSC as 3.95 Å (GluK2/K5em-AMPA – intact LBD), 3.82 Å (GluK2/K5em-AMPA – partially ruptured LBD) and 4.13 Å (GluK2/K5em-AMPA – partially ruptured 2 LBD). Local resolution was calculated using ResMap⁹² and visualized in UCSF ChimeraX⁹³.

For both the intact and partially ruptured LBD conformations of GluK2/K5em-5-IW-Kyn, the ATD layer and LBD-TMD assembly were independently refined, starting with particles from non-uniform refinement of the full-length receptor. For the ATD layers, the LBD-TMD signals were subtracted from the particles in Relion, then the particles were processed with ab initio reconstruction, heterogeneous refinement, and non-uniform refinement (C2 symmetry enforced for both) in cryoSPARC to the resolution of 3.11 Å (intact) and 3.03 Å (partially ruptured). For the LBD-TMD assemblies, the ATD signals were subtracted from the particles in Relion, then the particles were processed with ab initio reconstruction, heterogeneous refinement, and non-uniform refinement in cryoSPARC to the resolution of 3.65 Å (intact LBD, C2 symmetry) and 3.85 Å (partially ruptured LBD). C1 symmetry was used throughout unless otherwise noted.

For the intact and partially ruptured LBD conformations of GluK2/K5em-AMPA, the ATD layers and LBD-TMD assemblies were separately refined, starting with the particles used for non-uniform refinement of the full-length receptor. To refine the ATD layers, particles were subjected to signal subtraction to remove the LBD-TMD region in Relion, then processed with ab initio reconstruction and several rounds of heterogeneous refinements to isolate a uniform population of ATD particles. For the ATD of intact LBD conformation, non-uniform refinement (C2 symmetry) followed by local refinement was used to obtain an ATD structure at 3.32 Å resolution. Non-uniform refinement (C2 symmetry) was used to obtain ATD structures at 3.30 Å (partially ruptured state 1) and 3.52 Å (partially ruptured state 2) resolution. The LBD-TMD assemblies were resolved by subtracting ATD signals from the particles, followed by ab initio reconstruction, and several rounds of heterogeneous refinements to obtain a uniform particle subset, and then a final round of non-uniform refinements to result in a resolution of 4.39 Å (intact LBD, C2 symmetry), 4.22 Å (partially ruptured state 1), and 4.27 Å (partially ruptured state 2). The signal subtracted particles from LBD-TMD assemblies from both partially ruptured states were then pooled and underwent multiple rounds of ab initio and heterogeneous refinement to reveal four classes. C1 symmetry was used throughout unless otherwise noted.

We performed conformational variability analysis (3DVA)⁹⁴ in cryoSPARC for the full-length, ATD, and LBD-TMD of all the conformations of GluK2/K5em-5-IW-Kyn and GluK2/K5em-AMPA, and LBD-TMD of previously reported GluK2/K5em-CNQX dataset²³ with a mask that excluded solvent and micelle, with 3 variability components and a low-pass filter resolution of 5.5 Å. We identified distinct inter-domain LBD motion in the intact LBD state of GluK2/K5em-5-IW-Kyn compared to GluK2/K5em-CNQX.

Structural modeling and analysis

To model the intact and partially ruptured classes of GluK2/K5em-5-IW-Kyn, and the intact class of GluK2/K5em-AMPA, the ATD and LBD-TMD maps were first aligned to their respective full-length maps using Chimera⁹⁵. This ensured that ATD and LBD-TMD maps were correctly positioned relative to each other. The aligned ATD and LBD-TMD maps were then used for model building. Initial models for the intact and partially ruptured LBD classes of GluK2/K5em-5-IW-Kyn, and the intact class of GluK2/K5em-AMPA were built using the models of GluK2/K5em-CNQX (PDB 7KS0) and GluK2/K5em-L-Glu (PDB 7KS3). Then, the initial models were docked into the maps in Chimera, rebuilt using COOT⁹⁵ and refined using Phenix⁹⁶. Side chains were modelled where reliable electron density is observed; other residues were stubbed. Structural analysis was performed in ChimeraX. Unless otherwise noted, all distances are relative to Cα carbons of the noted residue. The pore radius of the TMD of the intact and partially ruptured classes of GluK2/K5em-5-IW-Kyn, and the intact class of GluK2/K5em-AMPA, were calculated using HOLE⁹⁷.

Functional patch-clamp characterization

Rat HA-SNAP-GluK2a and HA-SNAP-GluK5 variants were encoded on pRK5 plasmids¹⁰. Amino acid numbering includes the signal peptides. The R-edited GluK2 variant GluK2(Q621R) was used in all lifted whole-cell experiments to limit GluK2 expression levels and thereby increase GluK2/GluK5 heteromer expression. The Q-unedited GluK2 variant GluK2(Q621) was used for the 5-IW application experiments on outside-out patches (Fig. S7H,I). Otherwise, GluK2 corresponded to the I567V, Y571C-edited variant. GluK5 contained the known polymorphisms N887G, K889R, D910N. Amino acid exchanges were introduced using site-directed mutagenesis using polymerase chain reactions (PCRs) with oligonucleotides purchased from Sigma-Aldrich (Table S3). All coding sequences were verified by sequencing. These plasmids will be shared upon request from the Reiner Lab at Ruhr-University Bochum.

HEK293 and HEK293T cells were cultured in DMEM with 7 % FBS at 37 °C and 5% CO₂. Cells were transfected 24 h after initial seeding in plastic plates using 2.5 μl polyethylenimine (MW - 25,000, Sigma 408727, 1 μg/μl in water, pH 7) and a total of 0.3–0.6 μg DNA per ml medium. GluK2/GluK5 plasmids were transfected at a 1:5 ratio, and 0.05 μg pEGFP-N1 plasmid was added per well to serve as a transfection marker. To obtain isolated cells for lifted whole-cell experiments, the cells were reseeded 24–48 h after transfection by first washing the cells with DPBS and then rinsing them from the bottom using 2 times 1 ml enzyme-free cell dissociation solution (Sigma S-004). After centrifugation (3 min, 1500 g, room temperature), the supernatant was removed, and the cell pellet was resuspended in DMEM with 7 % FBS. The cell suspension was seeded onto glass coverslips weakly coated with poly-D-lysine (Sigma P6407, 1 μg/ml in DPBS). Measurements were performed 18–24 h after reseeding.

Patch-clamp measurements were performed on an inverted epifluorescence microscope (Leica DMI8) using an Axopatch 200B headstage/amplifier, a Digidata 1550A converter, Clampex 10.7 software (all Molecular Devices), and a micro-manipulator (Scientifica Patchstar). Patch pipettes were pulled from borosilicate glass capillaries and filled with intracellular solution (135 mM K-gluconate, 10 mM NaCl, 2 mM MgCl₂, 1 mM EGTA, 10 mM HEPES, pH 7.4). Measurements

were either performed in whole-cell configuration, with lifted whole cells, or outside-out patches, all in extracellular solution (138 mM NaCl, 1.5 mM KCl, 1.2 mM MgCl₂, 2.5 mM CaCl₂, 10 mM HEPES, pH 7.3). For whole-cell measurements on adherent cells, gravity-driven bath perfusion was used for ligand application. For fast ligand exchange^{24,98} lifted whole cells or outside-out patches were positioned either in front of a custom-made piezo-driven double-barreled borosilicate glass pipette (OD 2.0 mm, ID 1.4 mm, septum 0.2 mm, Warner Instruments) for the application of one ligand, or in front of a three-barreled glass pipette (2.0×2.0 mm, wall thickness 0.2 mm, channel dimensions 1.6×0.45 mm; CM Scientific, custom-made) for directly comparing the effects of two different ligands (see Fig. S1A). These perfusion pipettes were pulled using an electrode puller, their tip was broken off to yield an opening with a diameter of 200–400 μm, and mounted onto a piezo actuator (Physik Instrumente P842.20), which was driven by a voltage pulse (6 V ramp within 0.6 ms) and filtered with 2 kHz. The ligand and extracellular solutions were applied using a syringe pump and a speed of 0.15–0.3 ml/min per channel. Since GluK2/GluK5 heteromer currents were rather small, we mostly relied on lifted whole cells, which were lifted from the glass coverslips by applying slight negative pressure after break-in and moving the patch pipette slowly upwards. The agonists L-glutamate (Glu, Sigma 49449), (S)-5-iodowillardiine (5-IW, Tocris 0307) and (S)-AMPA (AMPA, Alomone Labs A-267) and the antagonist kynurenic acid (Kyn, Alomone Labs K-110) were applied in extracellular solution. Dose-response curves for GluK5 activation (Fig. S7F, G) were recorded in whole-cell configuration using a binding-impaired GluK2 subunit, GluK2(T690V)²⁴, which in combination with GluK5, results in non-desensitizing heteromer currents. GluK2(T690V) was co-expressed with GluK5(wt), GluK5(Y683I), or GluK5(N772E,773G) with EGFP as a transfection marker. Currents were elicited by gravity-driven bath perfusion of glutamate solutions (0.3–300 μM, ~4 ml/min). In the other whole-cell experiments (Fig. S10O, P) desensitization was blocked by 10 min incubation of the cells with 0.3 mg/ml Concanavalin A (ConA, Sigma L7647) in extracellular solution before the recordings. All recordings were performed in voltage-clamp mode with a holding potential of –70 mV, filtered with the Axopatch 200B 4-pole Bessel filter at 2 kHz and sampled at ≥ 5 kHz.

The outside-out patches and lifted whole cells were exposed to the ligands for 100–500 ms with an inter-sweep interval of 7–8 s to allow for recovery from desensitization. These ‘long’ pulses typically result in fast, ligand-induced desensitization. An extended 10 s application was performed to test for slow 5-IW-induced desensitization in Fig. S1I. To determine deactivation kinetics and estimate the heteromer content, a short glutamate pulse (0.6–1 ms) was applied to the cell, which was followed by a second pulse 200–1500 ms later, depending on the deactivation kinetics of the respective variant. These ‘short’ pulses are not long enough to desensitize all receptors but uncover their deactivation kinetics, which are particularly slow for GluK2/GluK5 heteromers¹⁸. To determine recovery from AMPA-induced desensitization (Fig. S10I,J), AMPA was applied for 500 ms to induce maximal desensitization, followed by a second 500 ms AMPA exposure (test pulse) at variable time intervals (5–500 ms). Here, the inter-sweep interval was 4 s.

Immunostaining for receptor cell surface expression

Cell surface expression levels of different GluK5 variants were analyzed by immunocytochemical staining against HA-tagged subunits (Fig. S7D,E). Glass coverslips were coated with PDL (5–10 μg/ml, Sigma P6407) and laminin (2.7 μg/ml, Sigma L2020) for 20 min at 37 °C. HEK293T cells were cultured and transfected as described above using GluK2 and the tested HA-SNAP-GluK5 plasmids in a 1:3 ratio. After 48 h expression, live cells were washed with 300 μl Krebs-Ringer's buffer (KRB; 125 mM NaCl, 4.8 mM KCl, 1.3 mM CaCl₂, 1.2 mM MgSO₄, 1.2 mM KH₂PO₄, 5.6 mM D-glucose, 25 mM HEPES, pH 7.3) and incubated with

monoclonal mouse anti-HA-tag antibody (2-2.2.14) (1:500, Thermo Scientific 26183) in KRB + 0.1% BSA at room temperature for 25 min. After two washing steps with KRB, cells were fixed for 5 min using 4% PFA in DPBS and washed twice with DPBS + 0.1% BSA. Secondary antibody staining with a goat anti-mouse AF488 antibody (1:500, Invitrogen A-10680) in DPBS + 0.1% BSA was performed for 30 min and the cells were then washed twice with DPBS + 0.1% BSA. Coverslips were mounted with Immu-mount (Epredia Shandon 9990402) and imaged on an inverted epifluorescence microscope (Leica DMI8) with a 20x/NA 0.40 objective (Leica HC PL FLUOTAR L CORR PHI). Green fluorescence was excited using a 470 nm LED and a 470/40 nm excitation filter in combination with a 495 nm dichroic mirror and a 525/50 nm emission filter. Images were taken from regions with good cell densities and no or few dead cells (absence of intracellular staining) using an EMCCD camera (Photometrics Evolve 512 delta) and a 5 s exposure time using Micro-Manager 2.0⁹⁹.

Analysis of patch-clamp measurements and receptor cell surface expression

Outside-out patches with 5-IW current responses lower than 5 pA and lifted whole cells with current responses lower than 20 pA were excluded from evaluation. Furthermore, recordings with >200 pA leak currents or an unstable baseline were excluded. Up to 10 sweeps were averaged for each cell. For display, electrical artifacts were reduced in amplitude. Peak currents, steady-state currents, rise times_{10–90%}, desensitization time constants τ_{des} , and deactivation time constants τ_{deac} were determined using Levenberg-Marquardt fitting with Clampfit 10.7 (Molecular Devices).

Desensitization kinetics τ_{des} were well approximated by single exponential fits, with the exception of GluK2/GluK5(Q684A) heteromers, which showed a minor slow phase, and GluK2/GluK5(N772E, N773G) heteromers. In both cases, weighted desensitization time constants are reported. For GluK2/GluK5 heteromers, the measured glutamate desensitization kinetics might also include contributions from homomeric receptors (see estimates of heteromer content). The ability to detect faster desensitization kinetics might have been limited by the relatively slow rise times of lifted whole cell experiments (glutamate current rise time_{10–90%} of 2.02 ± 0.21 ms, mean $\pm$ s.e.m). The deactivation kinetics were mostly fitted doubly exponentially, with the fast phase typically representing GluK2 homomer deactivation (3.14 ± 0.13 ms, mean $\pm$ s.e.m, see Fig. S1G, H) mixed with desensitizing current fractions, whereas the slow phase unequivocally represents GluK2/GluK5 heteromer deactivation (see e.g. Fig. 4E). To estimate the heteromer content in individual cells, we corrected the total current amplitude seen for the first glutamate application by subtracting the current loss seen with the second glutamate application in order to remove current fractions that desensitized rather than deactivated. The heteromer fraction is then given by dividing the current amplitude belonging to the slow deactivation phase by the corrected total amplitude. Most GluK5 variants yielded mid-to-high amounts of GluK2/GluK5 heteromers (cp. Table S2); cells with <25% heteromer content were excluded from the analysis. For the recovery experiment (Fig. S10I, J) the peak currents obtained from the test pulses were normalized to the first peak current of each individual sweep and then values from 3–5 subsequent runs of the recovery protocol were averaged. The figure shows averaged data without normalization (Fig. S10I). The normalized and averaged data points were then fitted with a single exponential using a Levenberg-Marquardt algorithm in OriginPro 2023 (Origin Lab Corporation).

Dose-response curves from individual cells were analyzed with a Hill equation ($h = 1$; $I = I_{max} * ([Glu]/(EC_{50} + [Glu]))$), normalized to their respective maximal amplitudes, and averaged (Fig. S7G). The reported EC₅₀ values reflect the averages obtained from individual fits.

Fluorescence images of immuno-stained cell surface receptors were analyzed with ImageJ/Fiji¹⁰⁰. The grayscale example image

(Fig. S7D) is shown with a low-intensity cut-off of 800 counts and a high-intensity cut-off of 5000 counts. For quantification, mean pixel intensities were obtained from individual, manually selected regions-of-interest (ROIs) that included the cell membrane of individual cells and that were corrected for the mean pixel intensity obtained from a nearby background region. Ten cells showing membrane fluorescence above background were selected from two regions of each coverslip. At least three coverslips from at least two independent transfections were analyzed per condition (Fig. S7E).

Statistical analysis was performed using OriginPro 2023. First, the data was tested for normal distribution with the Shapiro-Wilk test and Levene's test for equal variance ($p < 0.05$). Two groups were either compared using paired t -tests for data from the same cell, t -tests, or Welch's t -tests in case of unequal variance. Multiple groups (unequal variances) were assessed by a one-way Welch-ANOVA ($p < 0.05$) followed by a Games-Howell post hoc comparison to wildtype (wt) (Fig. S7B, C, E, I). Medians, means, and standard errors of the mean (s.e.m.) are reported, as indicated.

Sequence alignments

Local AMPA and kainate receptor sequence alignments (Fig. 4B–D) are based on the rat UniProt sequences GluK1 P22756-1, GluK2 P42260, GluK3 P42264-1, GluK4 Q01812, GluK5 Q63273, GluA1 P19490-2, GluA2 P19491-2, GluA3 P19492-2, and GluA4 P19493-2.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The atomic coordinates and cryo-EM density maps for intact and partially ruptured classes of GluK2/K5em-5-IW-Kyn and the intact class of GluK2/K5em-AMPA have been deposited in the PDB under accession codes [9OSF](#), [9OSG](#), and [9OSI](#), respectively, and in the Electron Microscopy Data Bank under accession codes [EMD-70805](#), [EMD-70806](#), and [EMD-70807](#), respectively. All other datasets generated during and/or analyzed during the current study are available in the Source Data file. Source data are provided with this paper.

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Author contributions

Purification, electron microscopy, image processing, and model building were performed by N.K.K., A.S., and J.M. Structural Analysis was performed by N.K.K., A.S., J.M., A.R., and J.L., with interpretation by all authors. Site-directed mutagenesis was performed by L.M.W. and S.L., functional measurements were performed and analyzed by L.M.W. AR and JL wrote the manuscript with input from all authors.

Competing interests

The authors declare no competing interests.

Additional information

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